

Mathematics A

Unit 2 • Chapter OL-T37 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

8 classified items • 11 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T37 • Expertise Q16+ • 8 items • 11 marks

Chapter	Items	Marks	Starts
01. OL-T37 Transformations of Graphs	8	11	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Transformations of Graphs

Unit 2 • Expertise • 11 marks • 8 item(s)

June 2024 | 4MA1-2H Q25 | 2 marks

Horizontal translation

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical scale or x-axis reflection

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical translation

June 2022 | 4MA1-2H Q23 | 1 marks

Horizontal scale or y-axis reflection

November 2021 | 4MA1-2H Q21 | 1 marks

Transform an explicit polynomial and simplify

November 2021 | 4MA1-2H Q21 | 1 marks

Translation vector applied to $y=f(x)$

November 2021 | 4MA1-2H Q21 | 1 marks

Describe from two curve equations

November 2021 | 4MA1-2H Q21 | 3 marks

Combined feature mapping

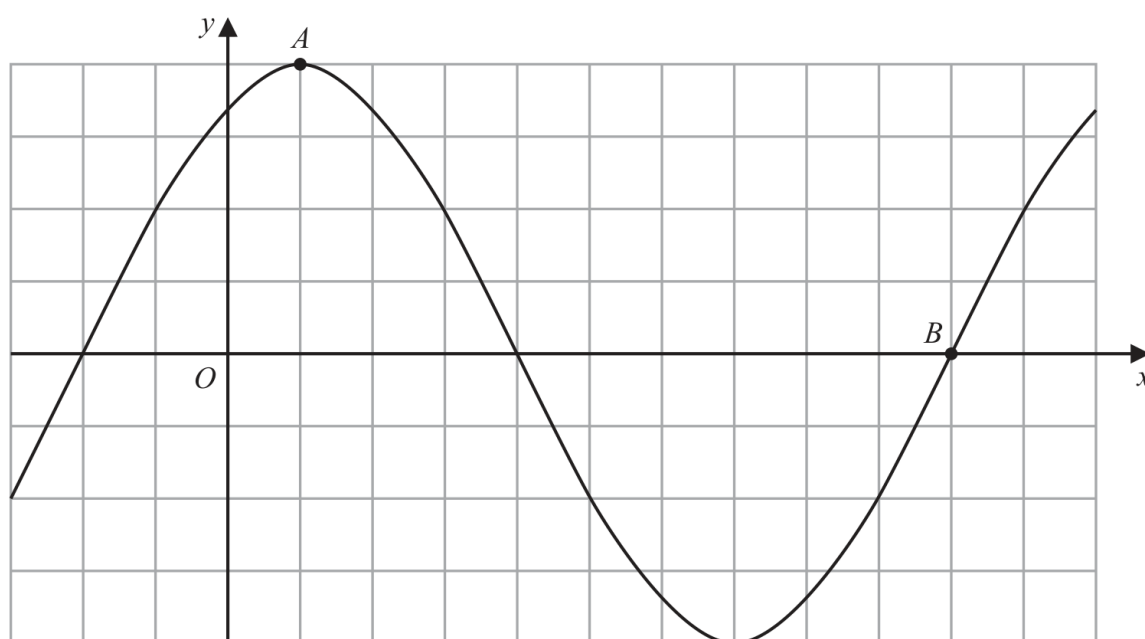
Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q25

2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

Continue your method**2 marks**

WORKING SPACE

4MA1-2H Q19 M02

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(.....,)
(1)

WORKING SPACE

4MA1-2H Q19 M03

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

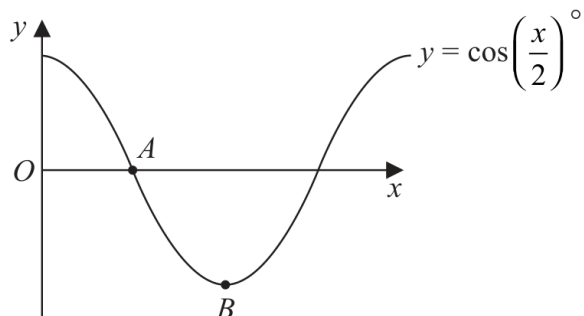
(.....,)
(1)

WORKING SPACE

4MA1-2H Q23 Q23(ii)

1 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(ii) Find the coordinates of the point B

(.....,)
(1)

Continue your method**1 marks**

WORKING SPACE

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

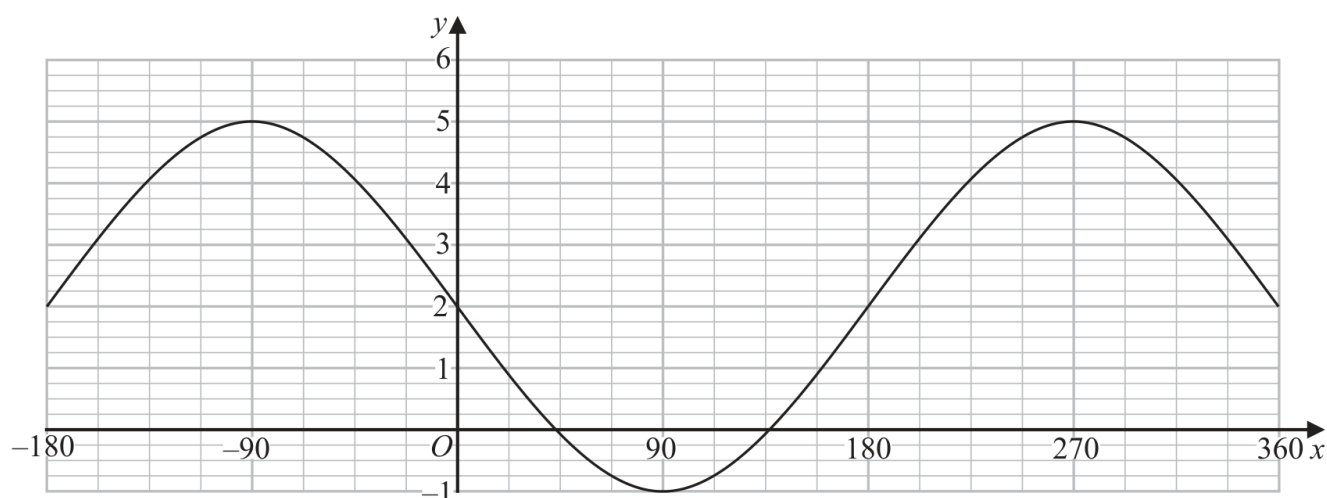
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(3)

YOUR WORKING

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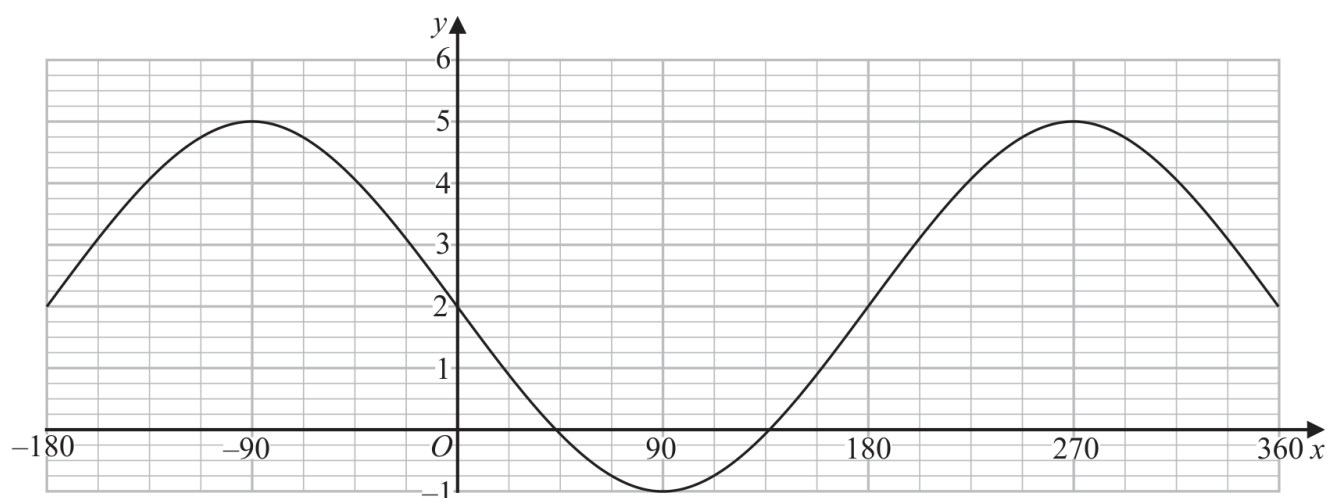
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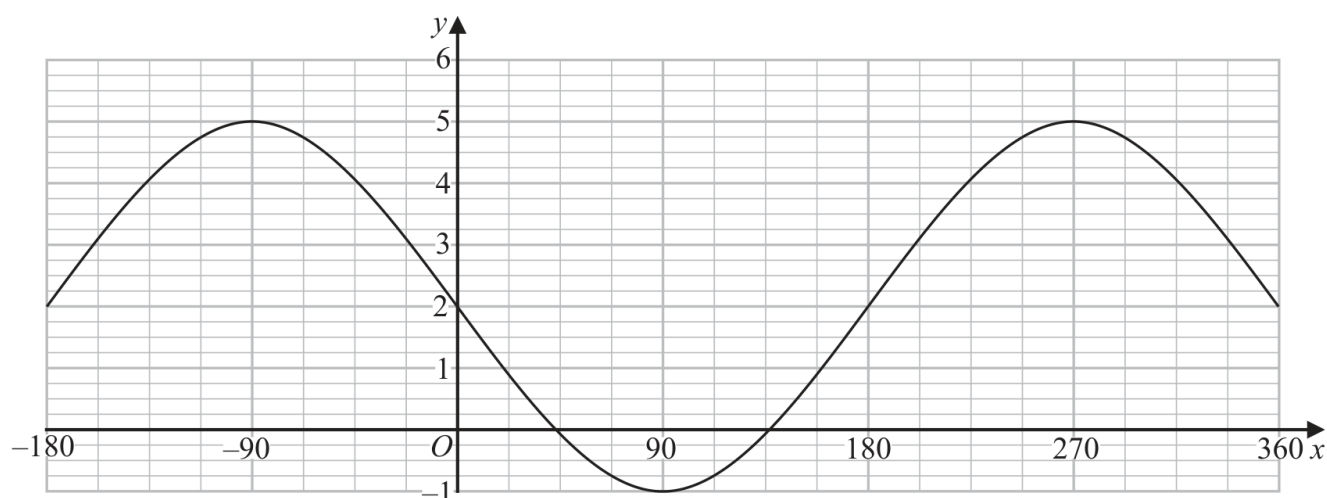
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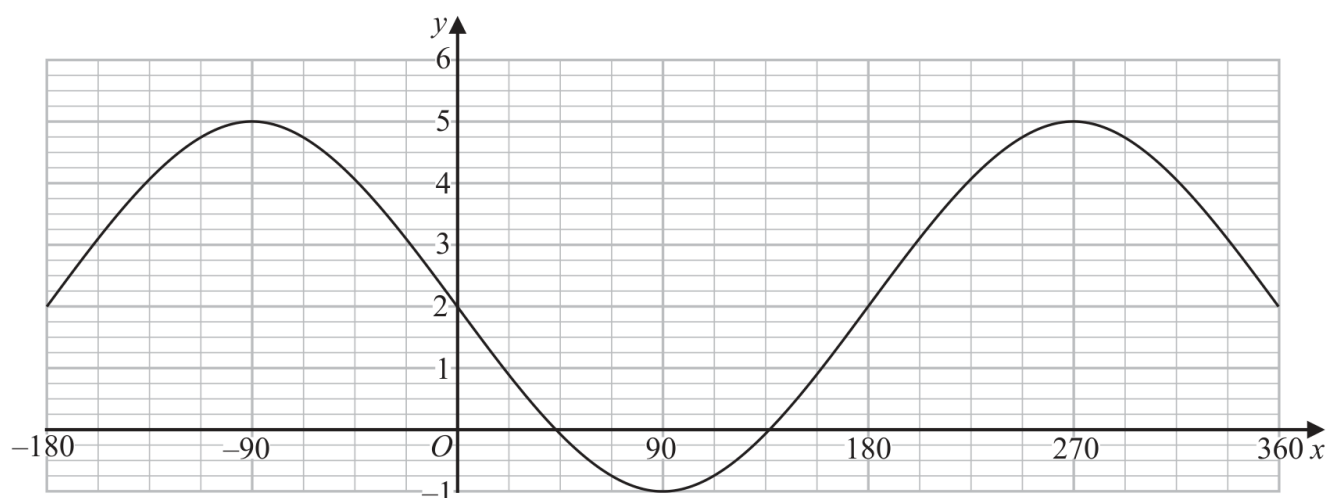
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